INCH-POUND

MIL-D-70566B (AR)
17 August 1993
SUPERSEDING
MIL-D-70566A
16 March 1987

MILITARY SPECIFICATION

DISTRIBUTION BOX ASSEMBLY: 9361136

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for the distribution box assembly, hereinafter referred to as the D-box, used in Gun, Anti-Aircraft, Artillery, Self-Propelled 20MM, M163A2, and Towed 20MM, M167A2 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1285

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SPECIFICATIONS

MILITARY

MIL-L-3661	- Lampholder, Indicator Lights, Indicator Light Housings, General Specification for
MIL-F-13926	- Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607	- Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-C-50576	- Circuit Board, Built-In-Test, Distribution Box (PIVADS): 9360925

STANDARDS

MILITARY

MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines
MS3106	- Connector, Plug Electric, Straight
MS3126	- Connector, Plug, Electrical, Crimp Type
MS3476	- Connector, Plug, Electric, Series 2, Crimp Type
MS24523	- Switch, Toggle, One Pole
MS24524	- Switch, Toggle, Two Pole
MS24525	- Switch, Toggle, Four Pole
MS25237	- Lamp, Incandescent, Single Contact

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the DODSSP - Customer Service, Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9361136	- Distribution Box Assembly
9361172	- Schematic Diagram, Electrical Distribution Box
9361175	- Wiring, Installation of, Distribution Box
9361217	- Remote BIT Tester Assembly
12950877	- Requirements for Soldering Electrical and Electronic Assemblies

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SPECIAL PACKAGING INSTRUCTION

PS 9361136 - Packaging of Distribution Box Assembly

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9361136, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the D-box shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.

3.3 Environmental service conditions.

3.3.1 Vibration. The D-box shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to the equivalent of 6000 miles of track-laying vehicle operational vibration as defined by MIL-STD-810 narrowband random-on-random data for equipment installed on the deck of the M113 armored personnel carrier.

3.3.2 Shock. The D-box shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10 percent, 11 milliseconds ± 10 percent, half-sine wave applied in each direction of three mutually perpendicular axes.

3.3.3 Altitude operating. The FCP shall meet the

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requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The D-box shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40°F and +155°F.

3.3.4.2 Non-operating (storage). The D-box shall not suffer permanent damage when subjected to the constant temperature extremes of -60°F for a period of 24 hours and +160°F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The D-box shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the D-box.

3.5 Performance.

3.5.1 Functional. The symbols referenced in the following paragraphs pertain to drawing 9361172.

3.5.1.1 Meters. The D-box shall contain meters to indicate the auxiliary power input voltage and current.

3.5.1.1.1 Voltmeter (M2). The voltmeter shall indicate zero to 50 Vdc.

3.5.1.1.2 Ammeter (M1). The ammeter shall indicate zero to 100 A. The meter shall be a 50 mV full scale type using an external shunt resistor.

3.5.1.2 Circuit breakers.

3.5.1.2.1 Fire control processor power circuit. The fire control processor (FCP) power circuit shall contain a 24 Vdc, 5A circuit breaker (CB1). It shall be possible to reset the circuit breaker from the front panel.

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3.5.1.2.2 Voltage converter power circuit. The voltage converter power circuit shall contain a 24Vdc, 10A circuit breaker (CB3). It shall be possible to reset the circuit breaker from the front panel.

3.5.1.2.3 System power circuit. The system power circuit shall contain a 24 Vdc, 15A circuit breaker (CB2). It shall be possible to reset the circuit breaker from the front panel.

3.5.1.2.4 Servo amplifier power circuits. The servo amplifier power circuits shall contain a 24 Vdc, 35A circuit breaker (CB6, CB7 and CB8) for each of the three servo amplifier power circuits. Overload in any of the three power circuits shall de-energize the turret power relay. The contacts of the circuit breakers shall automatically reset after removal of the overload.

3.5.1.3 NORM/STATIC/TEST switch (S1). This switch shall be a three position toggle switch located on the front panel. The switch shall be momentary in the TEST position. The switch shall control the TURRET POWER relay (K4) and the +24 Vdc circuit which feeds the three servo motor brakes in the gun system. Operation shall be as specified in Table I (NORM/STATIC/TEST switch operation).

TABLE I. NORM/STATIC/TEST switch operation.

POSITION	TURRET POWER RELAY (K4)	BRAKES
NORM	Coil Circuit CLOSED	Connected
STATIC	Coil Circuit OPEN	Disconnected
TEST	Coil Circuit CLOSED	Disconnected

3.5.1.4 Protective diodes. Diodes shall be provided in the power circuits to prevent the following:

a. Discharge of the system battery into the auxiliary power source (CR1).

b. Discharge of the gun battery into the auxiliary power source (CR2).

c. Damage to the 8.84 volt battery tap from application of voltages exceeding 8.84 Vdc (CR3).

d. Large voltage transients from the gun motor (CR4).

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e. Large voltage transients from the declutching solenoid (CR5).

f. Large voltage transients from the loading mirror (CR6).

3.5.1.5 Built-In-Test circuit board (A1A2). The D-box shall contain a Built-In-Test (BIT) 2 layer circuit board which monitors and transmits data on the gun battery voltage, high and low gun motor voltages, gun firing voltage, and specific switch and relay closures that control gun firing rates, clearing and braking. Data transmission shall be requested by the fire control processor. The D-box BIT circuit board (DBB) shall be as specified in MIL-C-70576.

3.5.1.6 Power control. Two power control relays shall be provided. These relays shall have 24 Vdc coils connected as specified in Table II (Power control relays).

TABLE II. Power control relays.

RELAY	COIL	COIL RETURN
SYSTEM POWER (K2)	J1-AC	+24 Vdc RETURN
RADAR POWER (K17)	J1-BE	+24 Vdc RETURN

3.5.1.6.1 SYSTEM POWER relay (K2). This relay shall be operated through control panel connector J2 and shall apply +24 Vdc system battery power to the control panel.

3.5.1.6.2 RADAR POWER relay (K17). This relay shall be operated through control panel connector J2 and shall supply +24 Vdc system battery power to radar connector J9.

3.5.1.7 Fire voltage. A fire voltage of +313.5 to +346.5 Vdc shall be provided by a CONVERTER (A3) operating from the 24 Vdc power. The fire voltage shall be delivered to the FIRING VOLTS relay (K6) contacts through a 24.9 ohm, 20 watt resistor. A shunt resistance of one megohm nominal shall be provided across the output of the converter.

3.5.1.8 Hi rate firing. With the connections of Figure 1, the RATE switch in the HI position and the GUN CLEAR switch in the AUTO position, the D-box shall perform as follows:

a. The SYSTEM POWER switch shall energize the SYSTEM POWER relay (K2) and apply power to the GUN POWER switch.

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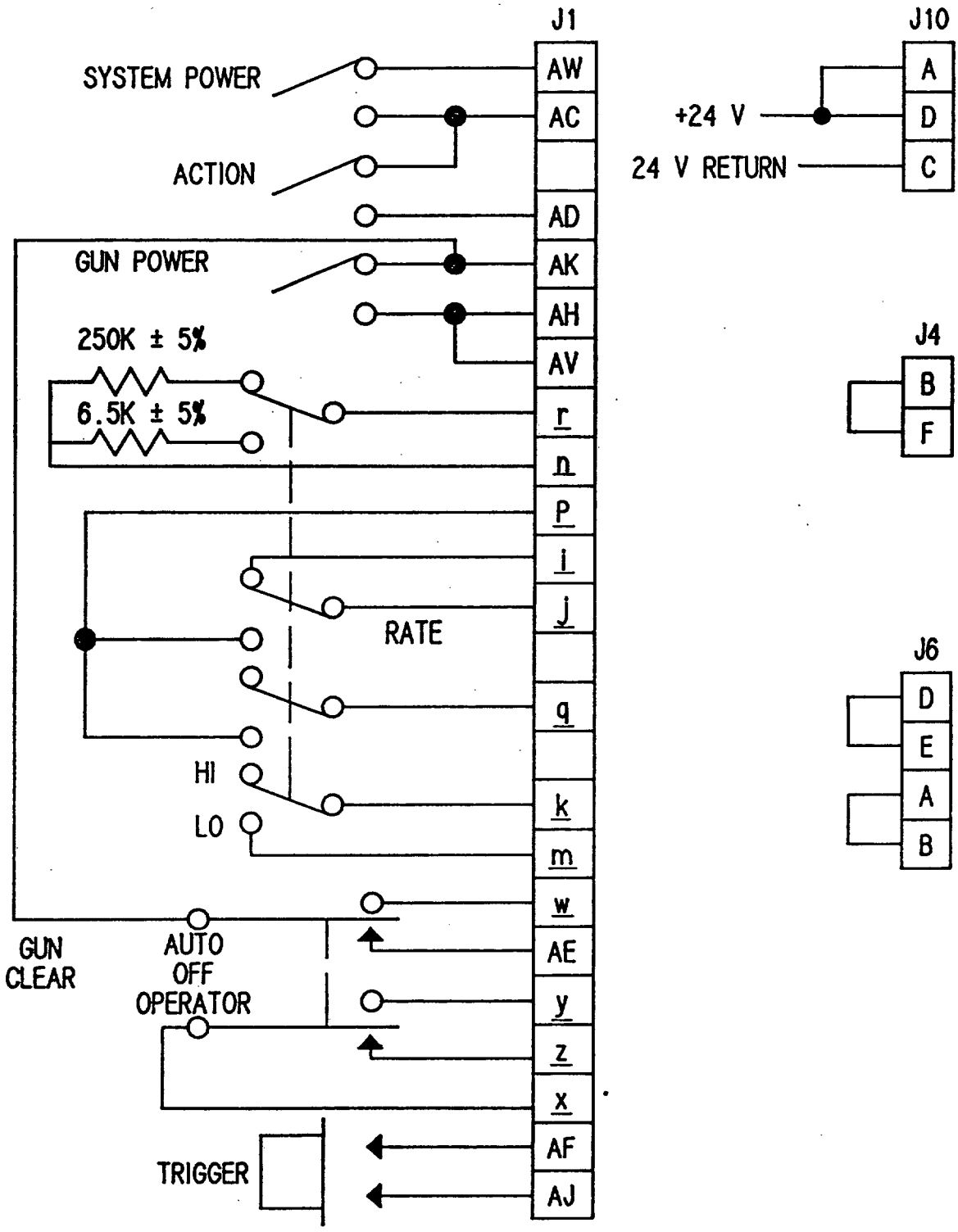


FIGURE 1. Functional connections (sheet 1 of 2).

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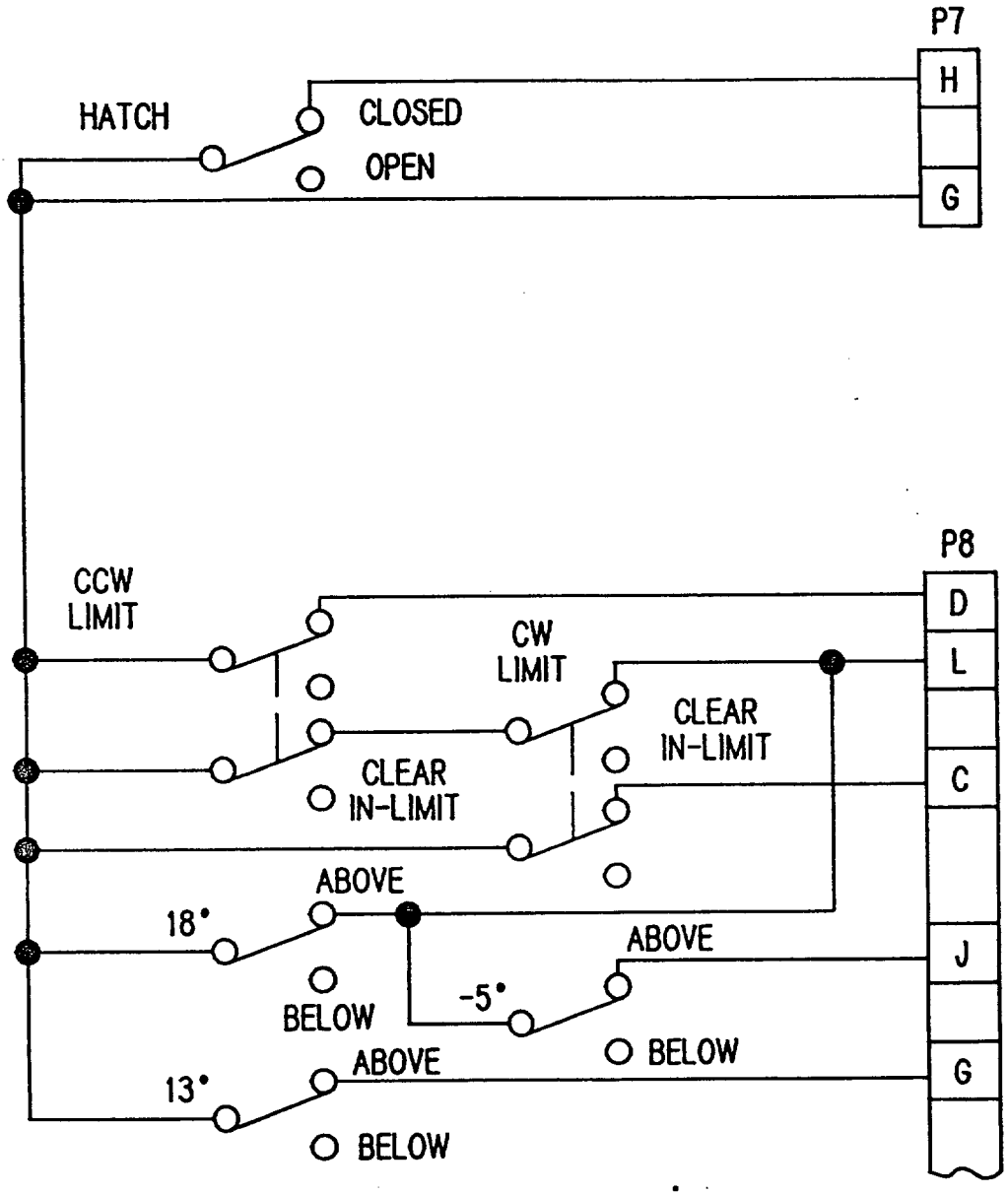


FIGURE 1. Functional connections (sheet 2 of 2).

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b. The ACTION SWITCH shall energize the ACTION relay (K3). The GUN POWER switch shall apply power to the fire voltage CONVERTER (A3) and through the contacts of the ACTION relay (K3) to the TRIGGER switch.

c. The TRIGGER switch shall energize the TRIGGER relay (K5). The TRIGGER relay (K5) shall energize the FIRING VOLTS (K6) and HI GUN MOTOR (K8) relays. After 0.87 to 2.1 seconds BURST LIMIT relay (K7) shall energize causing TRIGGER (K5) and HI GUN MOTOR (K8) to de-energize.

d. CLEARING relay (K11) shall energize for 1.0 to 2.23 seconds after TRIGGER relay (K5) de-energizes.

e. Releasing the TRIGGER switch shall de-energize BURST LIMIT relay (K7).

3.5.1.9 Lo rate firing. With the connections of Figure 1, the RATE switch in the LO position and the GUN CLEAR switch in the AUTO position, the D-box shall perform as follows:

a. The SYSTEM POWER switch shall energize the SYSTEM POWER relay (K2) and apply power to the GUN POWER switch.

b. The ACTION switch shall energize the ACTION relay (K3). The GUN POWER switch shall apply power to the fire voltage CONVERTER (A3) and through the contacts of the ACTION relay (K3) to the TRIGGER switch.

c. The TRIGGER switch shall energize the TRIGGER relay (K5). The TRIGGER relay (K5) shall energize the FIRING VOLTS (K6), HI GUN MOTOR (K8), and LO TIME DELAY (K9) relays. The LO TIME DELAY relay (K9) shall energize the LO GUN MOTOR relay (K10). After 26 to 75 milliseconds the BURST LIMIT relay (K7) shall energize causing the HI GUN MOTOR relay (K8) to de-energize.

d. When the TRIGGER switch is released the TRIGGER (K5) and BURST LIMIT (K7) relays shall de-energize immediately. The de-energized TRIGGER relay (K5) causes the CLEARING relay (K11) to energize. The LO TIME DELAY (K9) and LO GUN MOTOR (K10) relays shall de-energize 0.46 to 1.5 seconds after the TRIGGER relay (K5) has de-energized. CLEARING relay (K11) shall de-energize 1.0 to 2.23 seconds after the LO TIME DELAY relay (K9) de-energizes. FIRING VOLTS relay (K6) shal de-energize 0.7 to 1.95 seconds after the LO TIME DELAY relay (K9) de-energizes.

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3.5.1.10 Operator clearing. With the connections of Figure 1 and the SYSTEM POWER switch in the ON position and the GUN CLEAR switch in the OPERATOR position, the D-box shall perform as follows:

a. The ACTION switch shall energize the ACTION relay (K3), HI GUN MOTOR relay (K8), and CLEARING relay (K11).

b. Setting the ACTION switch to OFF shall de-energize the ACTION RELAY (K3) and HI GUN MOTOR relay (K8) immediately. The CLEARING relay (K11) shall de-energize in 0.7 to 1.95 seconds after the ACTION switch is set to OFF.

3.5.1.11 Hatch and elevation protective circuits. The D-box shall contain hatch and elevation protective circuits. These circuits shall be physically located in the D-box BIT circuit card. With the connections of Figure 1 and the SYSTEM POWER and ACTION switches ON, these circuits shall operate in conjunction with the hatch switch, the azimuth limit switches and the elevation limit switches to provide the logic in Table III (Hatch and elevation protective circuits logic table).

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TABLE III. Hatch and elevation protective circuits logic table.

LIMIT CIRCUIT CONDITION						
HATCH	-5 DEG	13 DEG	18 DEG	RIGHT	LEFT	LIMIT RELAYS DE-ENERGIZED
C	C	C	C	C	C	None
C	C	C	C	O	O	None
C	C	O	O	O	O	None
C	O	O	O	O	O	K13
O	O	O	O	O	C	K13 & K15
O	O	O	O	C	O	K13 & K16
O	C	O	O	C	C	None
O	C	C	O	C	O	K13
O	C	C	O	O	C	K13
O	C	C	O	C	C	None
O	C	O	O	C	O	K13 & K16
O	C	O	O	O	C	K13 & K15
O	C	O	O	O	O	K13, K15 & K16
O	C	C	O	O	O	K13
O	C	C	C	O	O	None

C = Switch Closed
O = Switch Open

3.5.1.12 Pulse generator. The D-box shall contain a capacitor of 1100 microfarads and a 49.9 ohm charging resistor connected through an external rounds counter switch.

3.5.1.13 Filter. The +24 Vdc supplied to pin AU of connector J1 shall contain an LC filter. The LC filter shall consist of a 2.08 millihenery inductor followed by a 100 microfarad capacitor.

3.6 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the requirements of note 2 of drawing 12950877.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as

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otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-F-13926 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50° to 90°F
b.	Relative humidity	90% RH max.
c.	Pressure	28 to 32 inches of Hg

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first

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article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies.

a. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAP's, the applicable quality assurance provisions of MIL-F-13926 shall apply.

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b. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.3). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provisions(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 shall be performed as specified in 4.4.2.3 and 4.5.

4.4.2.3 Functional tests. The tests in Table XI shall be performed on a 100 percent basis.

TABLE IV. Functional tests (100 percent inspection).

Characteristic	Requirement	Test Procedures
Critical	None	N/A
Major 101. Electrical 102. Performance	 3.4 3.5	 4.7.2 4.7.3
Minor	None	N/A

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4.5 Special sampling. One D-box out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table IV and shall then meet the tests in Table V. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE V. Special sampling.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
103. Vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105. Altitude	3.3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107. Humidity	3.3.5	4.7.1.5
Minor:	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of

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Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table VI (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

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TABLE VI. Test equipment list.

ITEM NUMBER		DESCRIPTION
1.	Power supply	Nominal value of 24 Vdc and 40 Vdc; 0-40 Vdc, adjustable voltage. 0-20 Adc, adjustable current limit. Load regulation 1% or less. Less than 5% ripple.
2.	Power supply	Nominal value of 5 Vdc; 0-10 Vdc adjustable voltage. Load regulation 1% or less from 10 to 50 milliamperes. Less than 5% ripple.
3.	Volt-ohmmeter	Dc voltage accuracy and resolution of 1% or better from 1 to 350 Vdc. Input impedance not less than 10 megohms when measuring 300 to 350 V. Resistance accuracy and resolution of 0.2% or better from one ohm to 1.2 Meg.
4.	Oscilloscope	Input impedance of one megohm or greater. Single sweep capability. Frequency range 0 to 1 MHz.
5.	Oscilloscope probe	Attenuation of 100X. Input resistance of 10 megohms.
6.	Audio oscillator (sinewave)	Capable of producing a frequency of 1000 Hz. Output impedance of 50 ohms. Output voltage into 50 ohms, not less than 3.0 volts rms.
7.	Test fixture	See Figure 2 for schematic and Table VII for the parts list.
8.	Remote BIT tester assembly	TS-3211/TSM-100B, NSN 4931-01-212-2436, Drawing 9361217 or equivalent.

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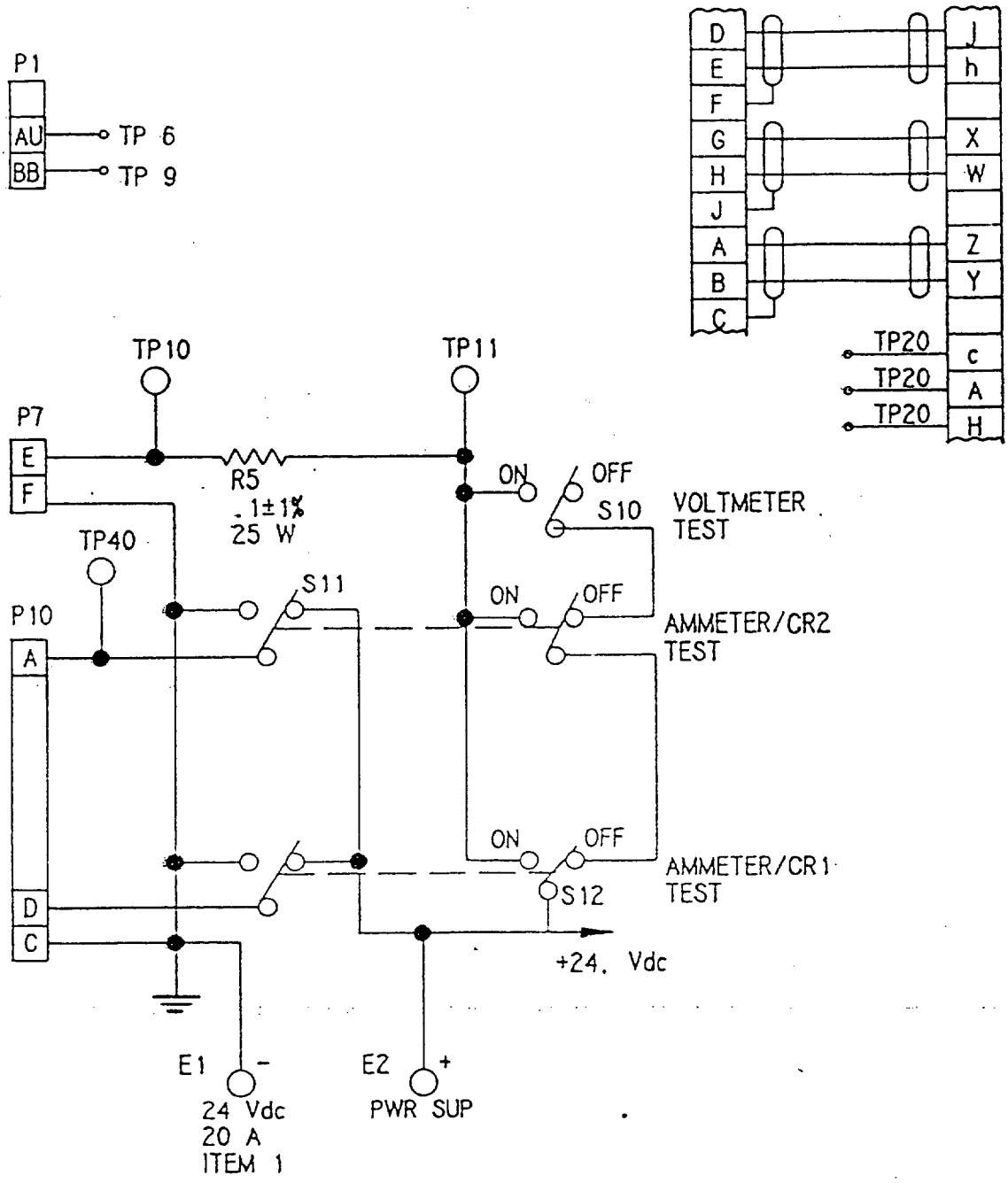
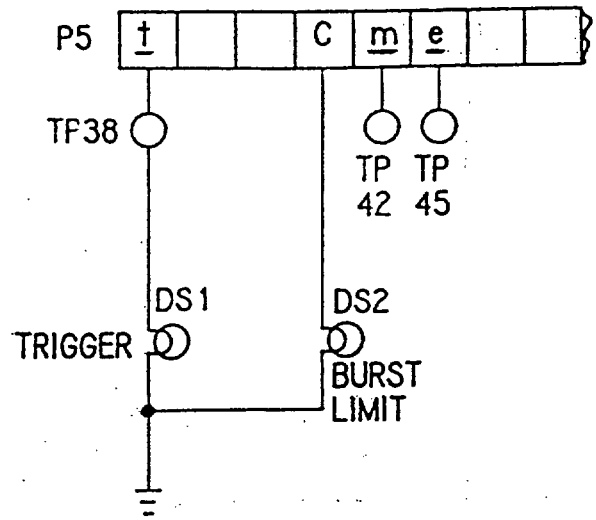


FIGURE 2. Test fixture (sheet 1 of 7).

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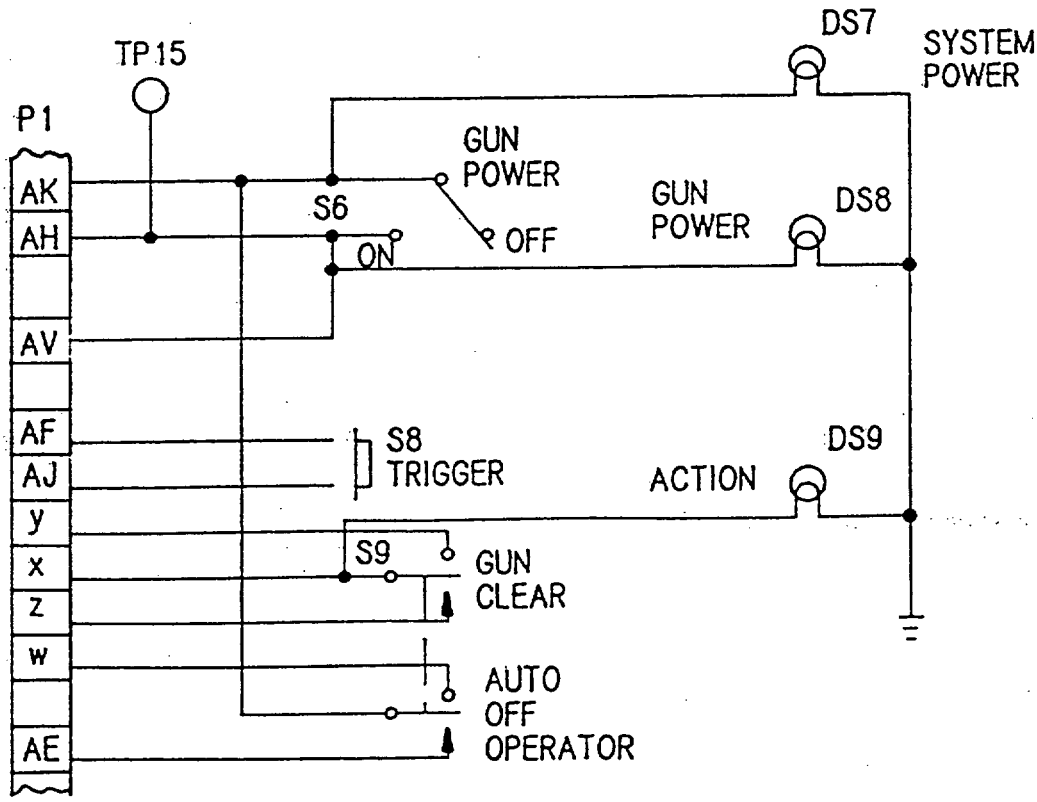


FIGURE 2. Test fixture (sheet 2 of 7).

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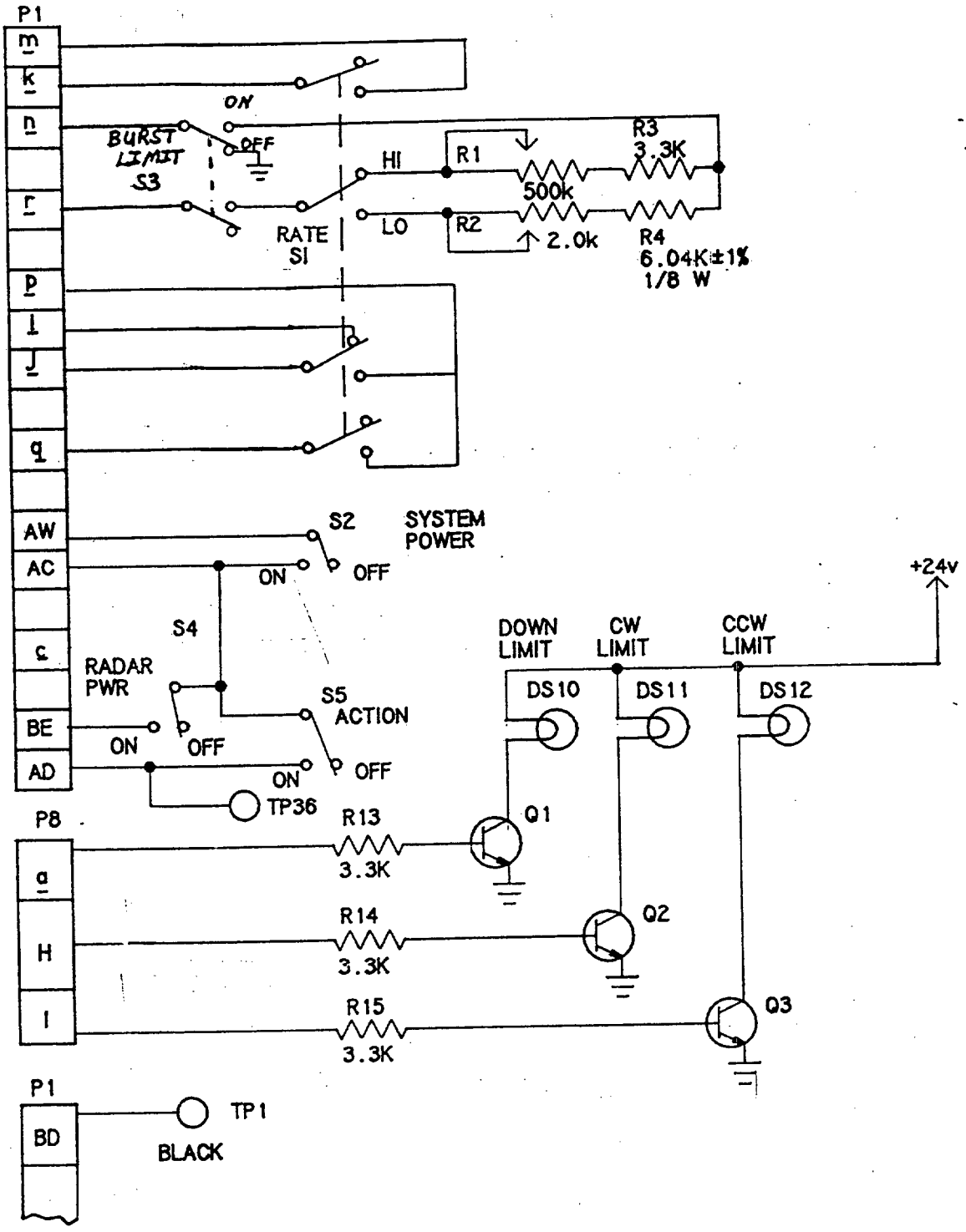


FIGURE 2. Test fixture (sheet 3 of 7).

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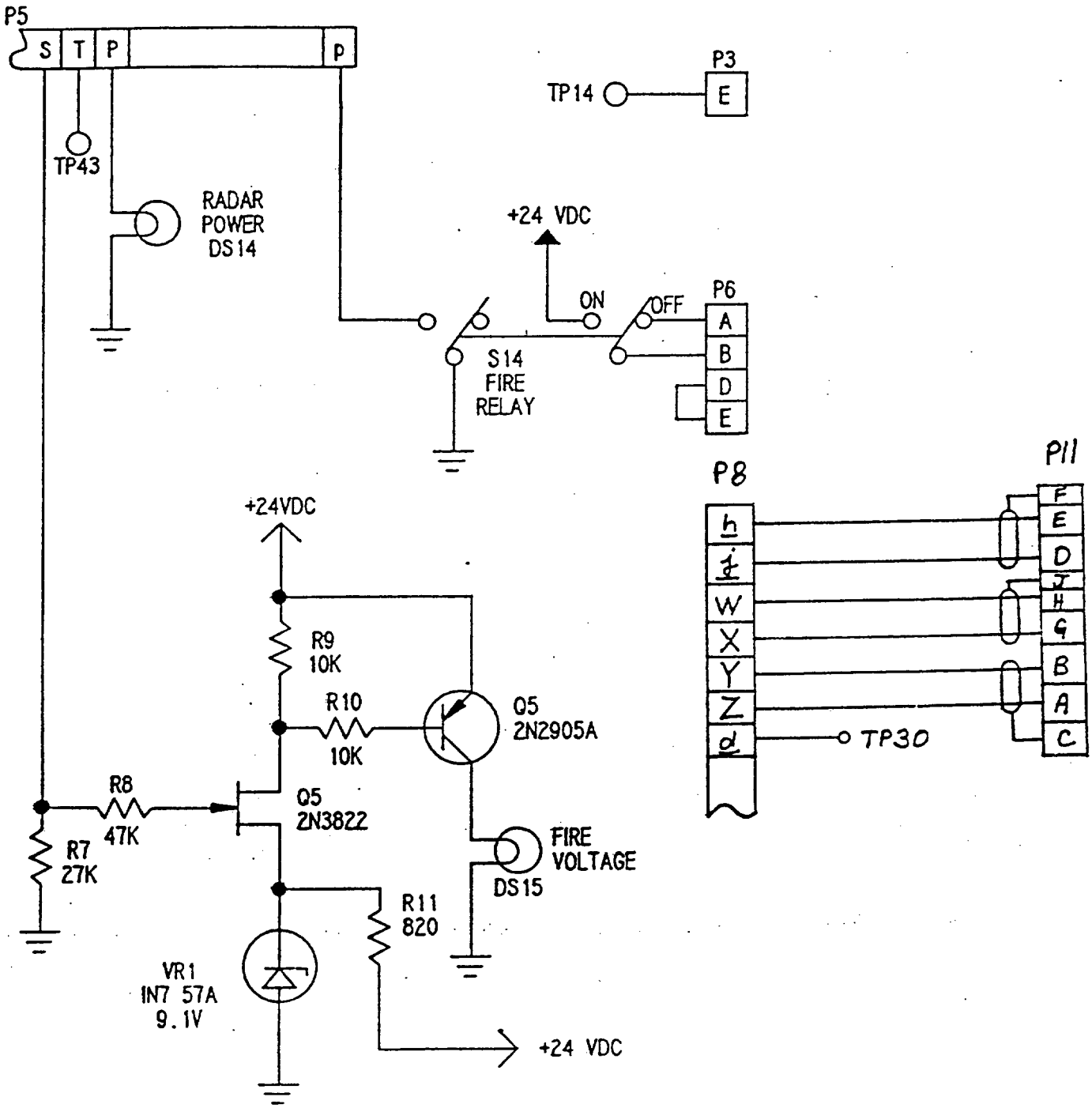


FIGURE 2. Test fixture (sheet 4 of 7).

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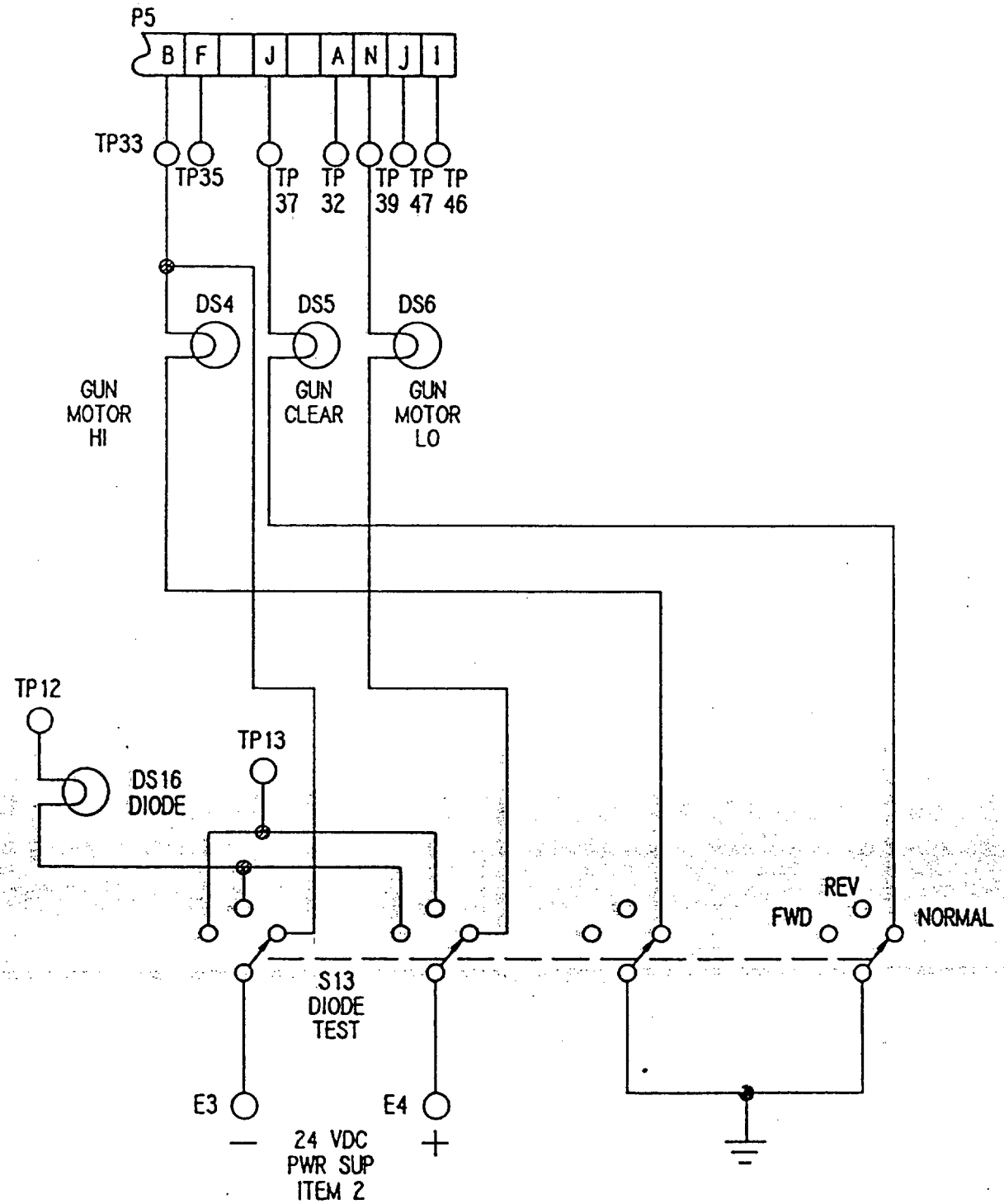


FIGURE 2. Test fixture (sheet 5 of 7).

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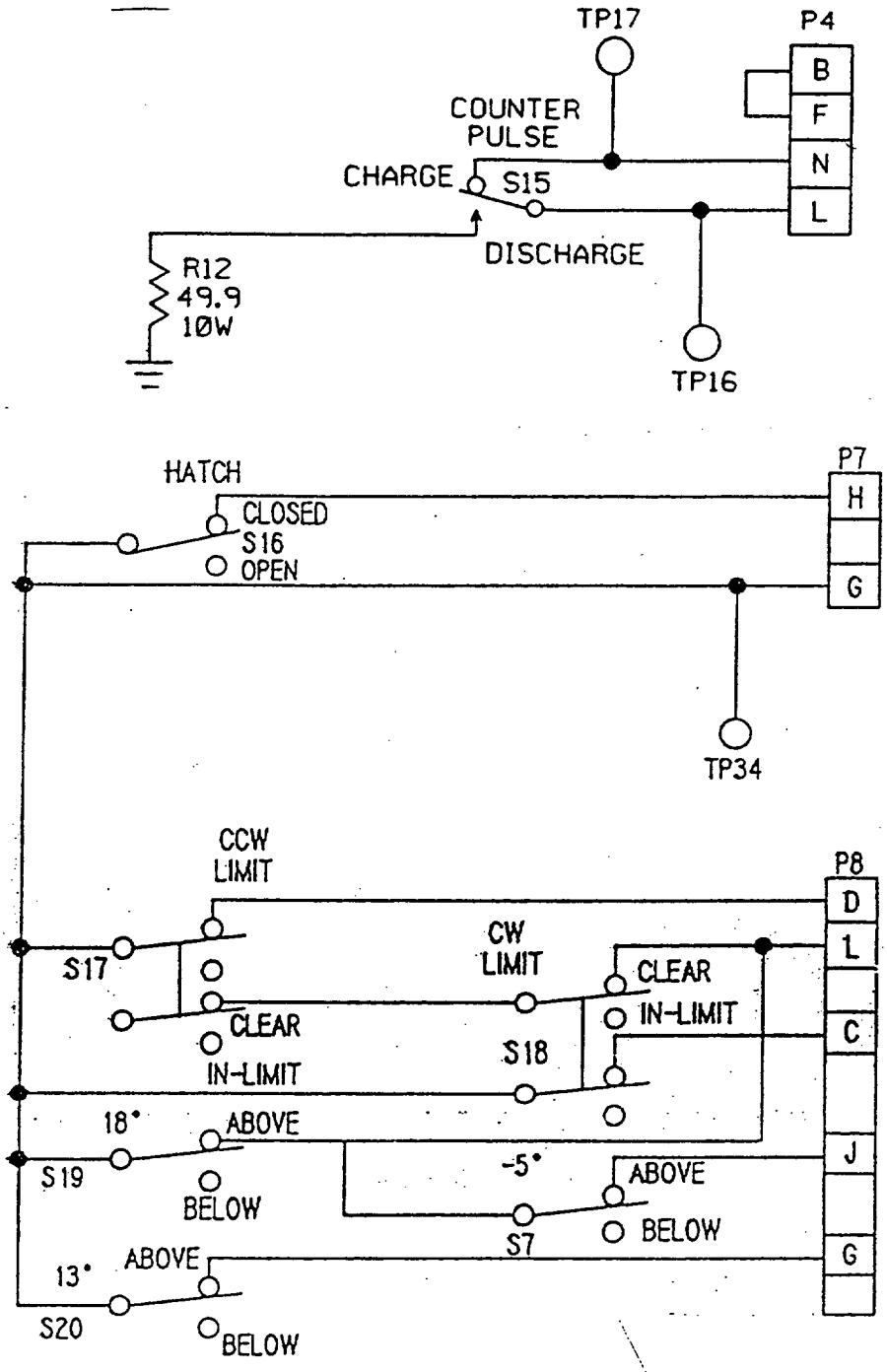
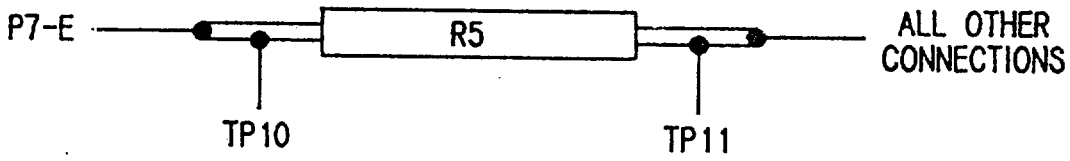


FIGURE 2. Test fixture (sheet 6 of 7).

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NOTES:

1. ADJUST R1 FOR A TOTAL RESISTANCE OF 250K ±5% ACROSS R1 AND R3.
2. ADJUST R2 FOR A TOTAL RESISTANCE OF 6.5K ±5% ACROSS R2 AND R4.
3. ALL RESISTORS ARE 1/4 WATT AND ±5% UNLESS OTHERWISE NOTED.
4. WIRE R5 AS SHOWN BELOW:



5. ALL WIRING TO P7. P10 AND THEIR ASSOCIATED COMPONENTS IS TO BE #12 AWG.
6. SWITCH S15 IS TO HAVE LOW CONTACT BOUNCE. A ROTARY TYPE IS RECOMMENDED.

FIGURE 2. Test fixture (sheet 7 of 7).

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TABLE VII. Test fixture parts.

DESIGNATION	DESCRIPTION	SUGGESTED PART NUMBER
DS1, DS2, DS4 thru DS12, DS14, DS15, DS16	Lamp miniature, 28 V, 0.04A	MS25237-327
P1	Connector, plug	MS3106F40-56P
P3	Connector, plug	NS3106A32-15P
P4	Connector, plug	MS3106F28-2P
P5	Connector, plug	MS3126F20-41P
P6	Connector, plug	MS3126F14-5P
P7	Connector, plug	MS3106F24-11S
P8	Connector, plug	MS3106F32-7P
P10	Connector, plug	MS3106F36-5S
P11	Connector, plug	MS3476W14-18P
Q1 thru Q3	Transistor, NPN, silicon	2N2219A
Q5	Transistor, field effort, N-channel	2N3822
Q6	Transistor, PNP, silicon	2N2905A
R1	Resistor, variable, 500K	RJR12FP504M
R2	Resistor, variable, 2.0K	RJR12FP202M
R3, R13, R14, R15	Resistor, 3.3K ± 5%, 1/4 watt	RCR07G332JM
R4	Resistor, 6.04K ± 1%, 1/8 watt	RNC55H6041FS
R5	Resistor, 0.1 ohm ± 1%, 25 watt	RE70GR100
R7	Resistor, 27K ± 5%, 1/4 watt	RCR07G273JM
R8	Resistor, 47K ± 5%, 1/4 watt	RCR07G473JM

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TABLE VII. Test fixture parts (continued).

DESIGNATION	DESCRIPTION	SUGGESTED PART NUMBER
R9 and R10	Resistor, 10K ± 5%, 1/4 watt	RCR07G103JM
R11	Resistor, 820 ± 5%, 1/4 watt	RCR07G821JM
R12	Resistor, 49.9 ± 5%, 10 watt	RWR78S49R9FM
S1	Switch, toggle, 4PDT, 28 Vdc, 1A	MS24525-23
S2, S4, S5, S6, S7, S10, S16, S19, S20	Switch, toggle, SPST, 28 Vdc, 5A	MS24523-22
S8	Switch, pushbutton, SPST, N.O. 28 Vdc, 5A	M8805/96-001
S9	Switch, toggle, 2 pole, 3 position, one momentary, 28 Vdc, 5A	MS24524-31
S11 and S12	Switch, toggle, DPDT, 28 Vdc, 20A	MS24524-23
S13	Switch, rotary, 4 pole, 3 position, non-shorting	M3786/13-0022
S3 and S14	Switch, toggle, DPDT, 28 Vdc, 5A	MS24524-23
S15	Switch, rotary, momentary non-shorting, 28 Vdc, 1.5A	Centralab 1451
S17 and S18	Switch, toggle, DPST, 28 Vdc, 5A	MS24524-22
TPs	Jacks, insulated, banana	HH Smith Type 1508
VR1	Diode voltage regulating, 9.1 V ± 5%	IN757A
XDS1, XDS2	Lampholder (MIL-L-3661/5A) White lens (MIL-L-3661/12)	LH73/1 LC12WT2

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TABLE VII. Test fixture parts (continued).

DESIGNATION	DESCRIPTION	SUGGESTED PART NUMBER
XDS4, XDS5, XDS6, XDS14	Lampholder (MIL-L-3661/5A) Green Lens (MIL-L-3661/12)	LH73/1 LC12GT2
XDS7, XDS8, XDS9, XDS15	Lampholder (MIL-L-3661/5A) Red Lens (MIL-L-3661/12)	LH73/1 LC12RT2
XDS10, XDS11, XDS12, XDS16	Lampholder (MIL-L-3661/5A) Yellow Lens (MIL-L-3661/12)	LH73/1 LC12YT2

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. This test shall be performed as specified in Method 514, Category 8, Procedure I of MIL-STD-810 and paragraph 3.3.1 herein. In step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The D-box shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the D-box shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post-test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10 percent amplitude and 11 ms ± 10 percent duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10 percent.

d. The shock measuring system shall have a flat frequency response within ± 10 percent from 20 to 2,000 Hertz.

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After the application of all the shock impulses, the D-box shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating). This test shall be performed as specified in Procedures II, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4.. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

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4.7.3 Performance. The selected D-box shall be tested as follows:

4.7.3.1 Condition of operation. Connect the test fixture (item 7, Table VI), the D-box under test, and connect power supplies (items 1 and 2, Table IV) to the test fixture, as shown in Figure 3. Connect the voltmeter (item 3, Table VI) and the oscilloscope (item 4, Table VI) to the test fixture as directed during the testing.

4.7.3.2 Preset controls. All controls are located on the test fixture (item 7, Table VI) and the power supplies (items 1 and 2, Table VI) with the exception of the NORMAL/STATIC/TEST switch (S1) which is on the D-box under test. Preset the NORMAL/STATIC/TEST switch to NORMAL. Preset all ON-OFF test set switches to OFF. Preset the remaining switches as listed below:

RATE to LO

DIODE TEST to NORMAL

HATCH to CLOSED

CCW LIMIT to CLEAR

CW LIMIT to IN LIMIT

18 deg to ABOVE

13 deg to ABOVE

-5 deg to ABOVE

GUN CLEAR to OFF

BURST LIMIT to ON

Insure that circuit breakers CB1 thru CB3 on the D-box are on. Turn on power supply (item 2, Table VI) and set the voltage to 24 ± 1 Vdc.

WARNING

TEST POINTS TP14 AND TP42 MAY HAVE HIGH VOLTAGE ON THEM AND COULD CAUSE SHOCK. USE WITH CAUTION.

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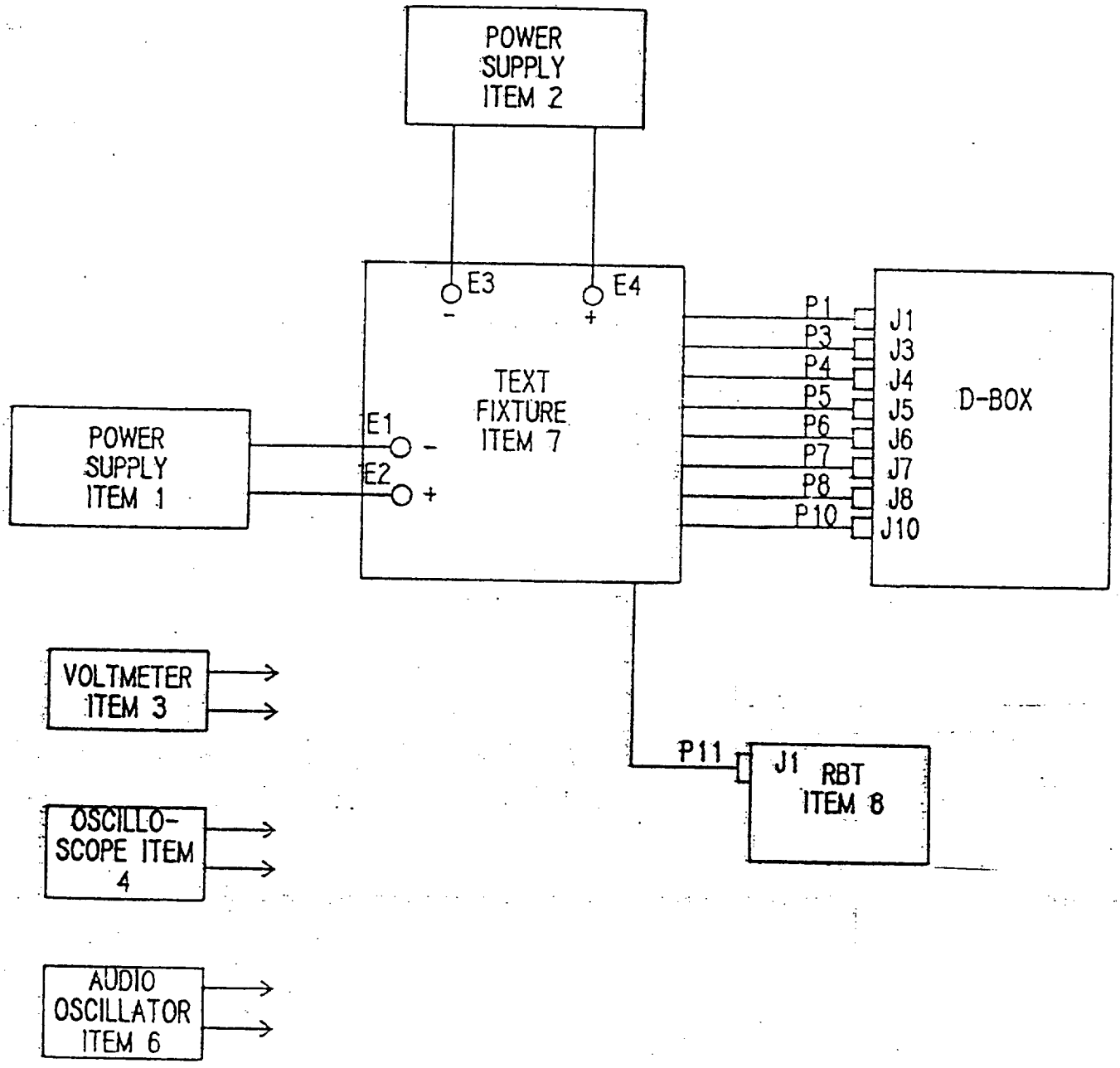


FIGURE 3. D-box test setup.

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4.7.3.3 Meters and diodes, CR1 and CR2. Connect the voltmeter (item 3, Table VI) to TP40 and the return to TP1. Turn on and adjust the power supply (item 1, Table VI) until the voltmeter indicates 40.0 ± 0.1 Vdc. Place the test set VOLTMETER TEST switch to ON. The D-box voltmeter (M2) must indicate 40 ± 1 volt. Adjust the power supply, item 1, until the voltmeter indicates 24.0 ± 0.1 Vdc. The D-box voltmeter (M2) must indicate 24 ± 1 volt. Operate the VOLTMETER TEST switch to OFF. Reduce the power supply (item 1, Table VI) voltage and current limit to zero. Connect the voltmeter (item 3, Table VI) to TP11 and the return to TP10. Perform the following steps:

a. Set the AMMETER/CR1 TEST switch to ON.

b. Adjust the power supply (item 1, Table VI) to 24 ± 1 Vdc. Slowly adjust the power supply current limit control for a reading of 2.0 ± 0.01 Vdc on voltmeter (item 3, Table VI).

c. The D-box ammeter (M1) must read 20 ± 1 amperes.

d. Set the AMMETER/CR1 TEST switch to OFF and set the AMMETER/CR2 TEST switch to ON.

e. Adjust power supply (item 1, Table VI) current limit control as necessary to maintain 2.0 ± 0.01 Vdc on the voltmeter (item 3, Table VI).

f. The D-box ammeter (M1) must read 20 ± 1 amperes.

g. Set the AMMETER/CR2 TEST switch to OFF.

4.6.3.4 Circuit breakers. Visually observe the front panel of the D-box and verify that the circuit breakers are marked in accordance with Table VIII (Circuit breakers).

TABLE VIII. Circuit breakers.

CIRCUIT BREAKER	CURRENT RATING (A)
FCP (CB1)	5
Voltage converter power (CB3)	15
System power (CB2)	15

4.7.3.5 NORMAL/STATIC/TEST switch (S1). Verify proper operation of the NORMAL/STATIC/TEST switch using the volt-ohmmeter (item 3, Table VI). Measure the resistance between test

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points specified in Table IX (NORMAL/STATIC/TEST switch). Continuity shall be indicated by a reading of less than 2.5 ohms.

TABLE IX. NORMAL/STATIC/TEST switch.

SWITCH POSITION	TP	TP	CONDITION
NORM	43	36	Continuity
STATIC	43	36	Open
TEST	43	36	Continuity
NORM	30	34	Continuity
STATIC	30	34	Open
TEST	30	34	Open

Return the NORMAL/STATIC/TEST switch (S1) to NORMAL.

4.7.3.6 Protective diodes. With the test point connections and DIODE TEST switch positions of Table X (Protective diodes), the DIODE indicator lamp condition must be as specified in the table.

TABLE X. Protective diodes.

DIODE	CONNECT TP	DIODE TEST SW	DIODE LAMP
CR3	32 to 12 and 39 to 13	Fwd	On
CR3	32 to 12 and 39 to 13	Rev	Off
CR4	45 to 12 and 33 to 13	Fwd	On
CR4	45 to 12 and 33 to 13	Rev	Off
CR5	45 to 12 and 37 to 13	Fwd	On
CR5	45 to 12 and 37 to 13	Rev	Off
CR6	45 to 12 and 46 to 13	Fwd	On
CR6	45 to 12 and 46 to 13	Rev	Off

Remove connections between test points. Return DIODE TEST switch to NORMAL.

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4.7.3.7 D-box BIT circuit. The DBB shall be tested with a volt-ohmmeter (item 3, Table VI) and remote BIT tester (item 8, Table VI) as follows:

a. On RBT, set the following switches in the sequence given:

SEND switch to the up position.

UNIT 4 TEST/NORMAL TEST switch to NORMAL TEST.

STATION 1234 switches to 0000.

ON/OFF power switch to ON and observe ON/OFF indicator illuminates.

Depress RESET LAMP TEST pushbutton switch and observe CHANNEL 1 and CHANNEL 2 DS lamps light.

b. Set SYSTEM POWER switch to ON and observe that SYSTEM POWER indicator illuminates.

c. Connect DVM to RBT J2(+) and J3(-).

d. On RBT perform the following thumbwheel switch setting in the sequence given:

STATION 1234 Switch	Normal Indication
8 0 0 0	8.51 to 8.91 Vdc
8 2 0 2	5.90 to 6.30 Vdc
8 6 0 6	5.90 to 6.30 Vdc
8 8 0 8	6.20 to 6.60 Vdc
8 10 0 10	8.20 to 8.90 Vdc
8 11 0 11	8.20 to 8.90 Vdc
8 14 0 14	8.20 to 8.90 Vdc
8 15 0 15	1.05 to 1.45 Vdc
9 0 1 0	1.05 to 1.45 Vdc

e. Set RADAR PWR switch to ON. On RBT, perform the following thumbwheel switch settings:

STATION 1234 Switch	Normal Indication
9 2 1 2	8.20 to 8.90 Vdc

f. Set RADAR PWR switch to OFF. On RBT, perform the following thumbwheel switch setting in the sequence given:

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STATION 1234 Switch	Normal Indication
9 4 1 4	8.20 to 8.90 Vdc
9 5 1 5	8.20 to 8.90 Vdc
9 7 1 7	8.20 to 8.90 Vdc
9 9 1 9	9.90 to 10.10 Vdc
9 10 1 10	9.90 to 10.10 Vdc
9 11 1 11	4.90 to 5.10 Vdc
9 15 1 15	8.51 to 8.91 Vdc

g. Set the ACTION switch to ON.

h. Perform the following in the sequence given:

Set GUN POWER switch to ON.

Set GUN CLEAR switch to AUTO.

Set RATE switch to HI.

Press and hold the TRIGGER switch to the ON position.

i. With the TRIGGER switch ON, perform the following RBT thumbwheel switch settings in the sequence given:

STATION 1234 Switch	Normal Indication
8 1 0 1	8.20 to 8.60 Vdc
8 5 0 5	8.20 to 8.60 Vdc
8 9 0 9	7.97 to 8.71 Vdc
8 11 0 11	8.20 to 8.60 Vdc
8 13 0 13	7.97 to 8.71 Vdc
9 3 1 3	8.20 to 8.60 Vdc

j. Set the BURST LIMIT switch to ON.

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k. With the TRIGGER switch ON, perform the following RBT thumbwheel switch setting in the sequence given:

STATION 1234 Switch	Normal Indication
8 3 0 3	8.20 to 8.60 Vdc
9 8 1 8	8.20 to 8.60 Vdc

l. Set the TRIGGER and BURST LIMIT switches to the OFF position.

m. Set RATE switch to LO. Press and hold TRIGGER switch to ON.

n. With the TRIGGER switch ON, perform the following RBT thumbwheel setting:

STATION 1234 Switch	Normal Indication
8 12 0 12	8.20 to 8.90 Vdc

o. Release the TRIGGER switch.

p. On RBT, perform the following thumbwheel setting:

STATION 1234 Switch	Normal Indication
9 1 1 1	8.20 to 8.60 Vdc

q. Press and hold GUN CLEAR switch to OPERATOR position.

NOTE

GUN CLEAR switch must have remained in OPERATOR position at least 3 seconds before performing the next step.

r. On RBT, perform the following thumbwheel setting:

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STATION 1234 Switch	Normal Indication
8 7 0 7	8.20 to 8.60 Vdc

s. Set the GUN CLEAR switch to OFF.

t. Set all ON/OFF switches on the TEST FIXTURE to OFF. Set AMMETER/CR1 TEST switch to ON. Set SYSTEM POWER switch to ON.

u. On RBT, perform the following thumbwheel setting:

STATION 1234 Switch	Normal Indication
9 6 1 6	8.20 to 8.60 Vdc

v. Set all ON/OFF switches to OFF.

w. Set BURST LIMIT switch to ON.

4.7.3.8 Power control. With the SYSTEM POWER switch ON, the SYSTEM POWER indicator must illuminate. Set the RADAR PWR switch to ON. The RADAR POWER indicator must illuminate. Return the RADAR PWR switch to OFF.

4.7.3.9 Fire voltage. Connect the ohmmeter (item 3, TAble VI) between TP45 and TP42. The ohmmeter reading must not be greater than 1.2 megohms. Set the FIRE RELAY switch to ON. Connect the ohmmeter (item 3, Table VI) to TP42 and TP14. The ohmmeter must indicate 24.9 $\pm$ 1 ohms. Return the FIRE RELAY switch to OFF and remove the ohmmeter leads. Connect the voltmeter (item 3, Table VI) between TP42 (High) and TP45 (Low). Operate the GUN POWER switch to ON. The voltmeter must indicate +313.5 to +346.5 Vdc. Return the GUN POWER switch to OFF.

4.7.3.10 Functional circuits.

4.7.3.10.1 Sequence of operation. Perform the following steps:

a. ACTION switch to ON. The ACTION indicator must light.

b. Depress the TRIGGER switch momentarily. The GUN MOTOR HI, GUN MOTOR LO and FIRE VOLTAGE indicators must remain OFF.

c. RATE switch to HI and repeat step b.

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d. GUN POWER switch to ON. The GUN POWER indicator must light.

e. ACTION switch to OFF.

f. GUN CLEAR switch to OPERATOR and hold.

g. ACTION switch to ON. The GUN MOTOR HI, ACTION and GUN CLEAR indicator must light. The FIRE VOLTAGE indicator must remain OFF.

h. Release GUN CLEAR switch.

i. RATE switch to LO.

j. GUN CLEAR switch to OPERATOR and hold. The GUN MOTOR HI, ACTION and GUN CLEAR indicators must light. The FIRE VOLTAGE indicator must remain OFF.

k. GUN CLEAR switch to AUTO.

l. Depress the TRIGGER switch and hold. The TRIGGER indicator, GUN MOTOR LO indicator and FIRE VOLTAGE indicator must light. The GUN MOTOT HI indicator must come on momentarily. The BURST LIMIT indicator must light after a short delay.

m. Release the TRIGGER switch. The GUN CLEAR indicator must come on momentarily.

n. Set the RATE switch to HI.

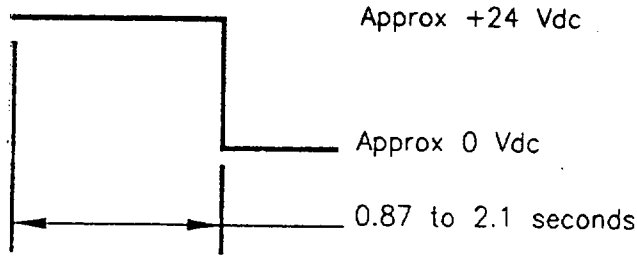
o. Depress the TRIGGER switch and hold. The GUN MOTOR HI, TRIGGER and FIRE VOLTAGE indicators must light briefly then go out. When the GUN MOTOR HI indicator goes out, the GUN CLEAR and BURST LIMIT indicators must light. The GUN CLEAR indicator must go out after a delay.

p. Release the TRIGGER switch.

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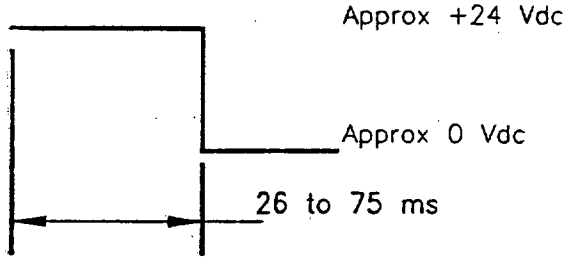
4.7.3.10.2 Time delays.

4.7.3.10.2.1 Burst limit. Connect the oscilloscope (item 4, Table VI) axis to TP35. Set up the oscilloscope sweep for single sweep, external positive triggering. Connect the external trigger to TP38. Depress and hold the TRIGGER switch. The oscilloscope must display the waveform shown below. Release the TRIGGER switch.



The waveform may be repeated by depressing and holding the TRIGGER switch. This sequence may be repeated as often as necessary.

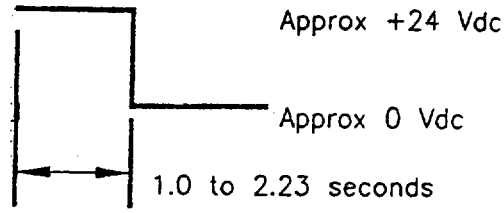
4.7.3.10.2.2 Gun motor high. Set the RATE switch to LO. Depress and hold the TRIGGER switch. The oscilloscope must display the waveform shown below. Release the TRIGGER switch.



The waveform may be repeated by depressing and holding the TRIGGER switch. This sequence may be repeated as often as necessary.

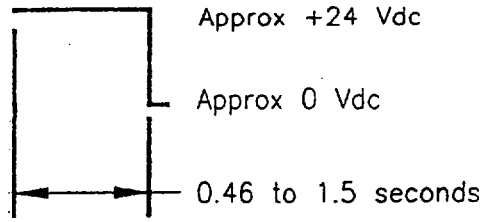
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4.7.3.10.2.3 Clear. Connect the oscilloscope y-axis to TP37. Set up the oscilloscope sweep for single sweep, external negative trigger. Connect the external trigger to TP47. Depress and release the TRIGGER switch. The oscilloscope must display the waveform shown below:



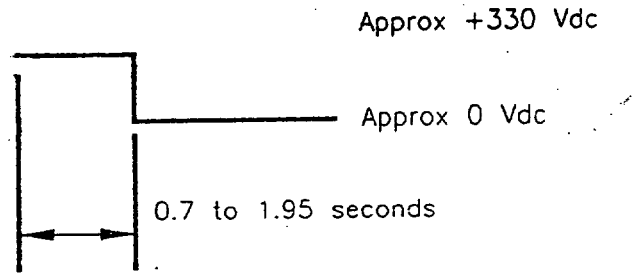
The waveform may be repeated by depressing and releasing the TRIGGER switch. This sequence may be repeated as often as necessary.

4.7.3.10.2.4 LO rate. Connect the oscilloscope y-axis to TP47. Connect the external trigger to TP38. Depress and release the TRIGGER switch. oscilloscope must display the waveform shown below:



The waveform may be repeated by depressing and releasing the TRIGGER switch.

4.7.3.10.2.5 Fire voltage. Connect the oscilloscope y-axis to TP14 using a 100X probe (item 5, Table VI). Connect the external trigger to TP47. Depress and release the TRIGGER switch. The oscilloscope must display the waveform shown below:



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The wave form may be repeated by depressing and releasing the TRIGGER switch as often as necessary.

Remove the oscilloscope connections.

4.7.3.10.3 Hatch and elevation protection circuits. Perform the following steps:

a. Place CW LIMIT switch to CLEAR. Observe that the CCW LIMIT, CW LIMIT and DOWN LIMIT indicators are OFF.

b. Set the CCW LIMIT and CW LIMIT switches to the IN LIMIT position. The CCW LIMIT, CW LIMIT and DOWN LIMIT indicators shall remain OFF.

c. Set the 13 deg and 18 deg switches to the BELOW position. The CCW LIMIT, CW LIMIT and DOWN LIMIT indicators shall remain OFF.

d. Set the -5 deg switch to the BELOW position. The CCW LIMIT and CW LIMIT indicators shall remain OFF, and the DOWN LIMIT indicator is ON.

e. Set the HATCH switch to OPEN and CCW LIMIT switch to CLEAR. Observe that only the CCW LIMIT indicator is OFF.

f. Set the CCW LIMIT switch to IN LIMIT and CW LIMIT switch to CLEAR. Observe that the DOWN and CCW LIMIT indicators are ON, and CW LIMIT indicator is OFF.

g. Set the -5 deg switch to ABOVE and CCW LIMIT switch to CLEAR. Observe that the CCW LIMIT, CW LIMIT and DOWN LIMIT indicators are OFF.

h. Set the 13 deg switch to ABOVE and CCW LIMIT switch to IN LIMIT. Observe that the CCW LIMIT and CW LIMIT indicators are OFF, and DOWN LIMIT is ON.

i. Set the CCW LIMIT switch to CLEAR and CW LIMIT switch to IN LIMIT. Observe that the CCW LIMIT and CW LIMIT indicators remain OFF, and DOWN LIMIT remains ON.

j. Set the CW LIMIT switch to the CLEAR position. Observe that the CCW LIMIT, CW LIMIT and DOWN LIMIT indicators are OFF.

k. Set the 13 deg switch to BELOW and CCW LIMIT switch to IN LIMIT positions. Observe that only the CW LIMIT indicator is OFF.

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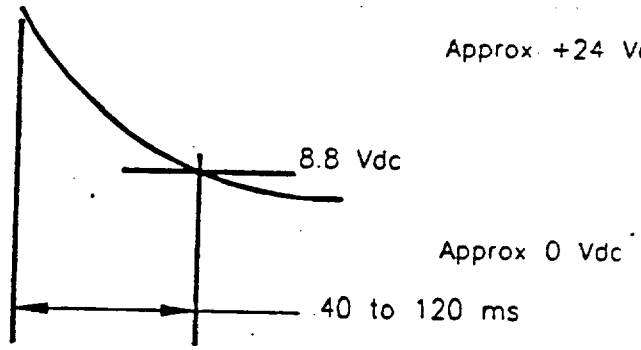
l. Set the CCW LIMIT switch to CLEAR and the CW LIMIT switch to IN LIMIT. Observe that only the CCW LIMIT indicator is OFF.

m. Set the CCW LIMIT switch to the IN LIMIT position and observe that all indicators are ON.

n. Set the 13 deg switch on the ABOVE position and observe that the CCW LIMIT and CW LIMIT indicators are OFF.

o. Set the 18 deg switch to the ABOVE position and observe that the CCW LIMIT, CW LIMIT and DOWN LIMIT indicators are OFF.

4.7.3.11 Pulse generation. Connect the oscilloscope (item 4, Table VI) y-axis to TP16. Set up the oscilloscope sweep for single sweep, internal positive triggering. Operate the COUNTER PULSE switch and observe the following waveform on the oscilloscope.



The waveform may be repeated by releasing and depressing the COUNTER PULSE switch as often as necessary. Pause briefly in the CHARGE position to provide time for the capacitor in the D-box to fully charge.

Turn off the SYSTEM POWER switch. Using the volt-ohmmeter (item 3, Table VI) measure the resistance between TP15 and TP17. The ohmmeter shall indicate 49.9 ± 0.5 ohms.

4.7.3.12 Filter. Connect the 50 ohm output of the audio oscillator (item 6, Table VI) to TP9 with the return to TP1. Also, connect the y-axis of the oscilloscope (item 4, Table VI) to TP9. Adjust the audio oscillator for 1.0 KHz. Adjust the scope sweep to display the 1.0 KHz waveform. Set the audio oscillator output to provide 1.0 volt peak-to-peak as read on the oscilloscope. Move the oscilloscope y-axis connection to TP6 and observe that the oscilloscope indicates a peak-to-peak waveform of 0.09 to 0.2 volts.

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4.8 Inspection equipment. The inspection equipment required to perform the inspections specified herein is identified in the "Inspection Method Reference" column of the Classification of Characteristics listings starting with 4.4.2.1. Contractor inspection equipment designs shall be submitted for Government approval as specified in the contract. Designs which provide variable measurements instead of attributes data are preferred in order to facilitate the use of statistical process control. See MIL-A-48078 and 6.2 herein.

4.9 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of Note 2 of Drawing 12950877 as specified in paragraph 3.7.

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Special Packaging Instruction PS 9361136. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The D-box shall be used in the Gun, Anti-Aircraft, Artillery, VADS self-propelled M163A2 and towed M167A2 vehicles.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1).

c. Requirements for submission of first article sample.

d. Applicable stock number.

e. Packaging requirements, if other than specified in Section 5.

f. Serialization requirements, if applicable.

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g. Certificate of conformance for each lot or shipment of product.

6.3 Submission by the contractor.

a) Inspection equipment designs for approval. Submit copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

b) Alternative quality conformance provisions for approval. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation to the technical activity responsible for the preparation of this specification.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Fire Control System
Gun, anti-aircraft artillery
M163A2 Self-Propelled 20MM
M167A2 Towed 20MM
Product Improved Vulcan Air Defense System (PIVADS)

6.6 Changes from previous issue. The previous issues were for ARDEC internal use only. They were not forwarded to Naval Publications and Forms Center (NPFC) for publication. This issue is initial public release for unlimited distribution.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A744)

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